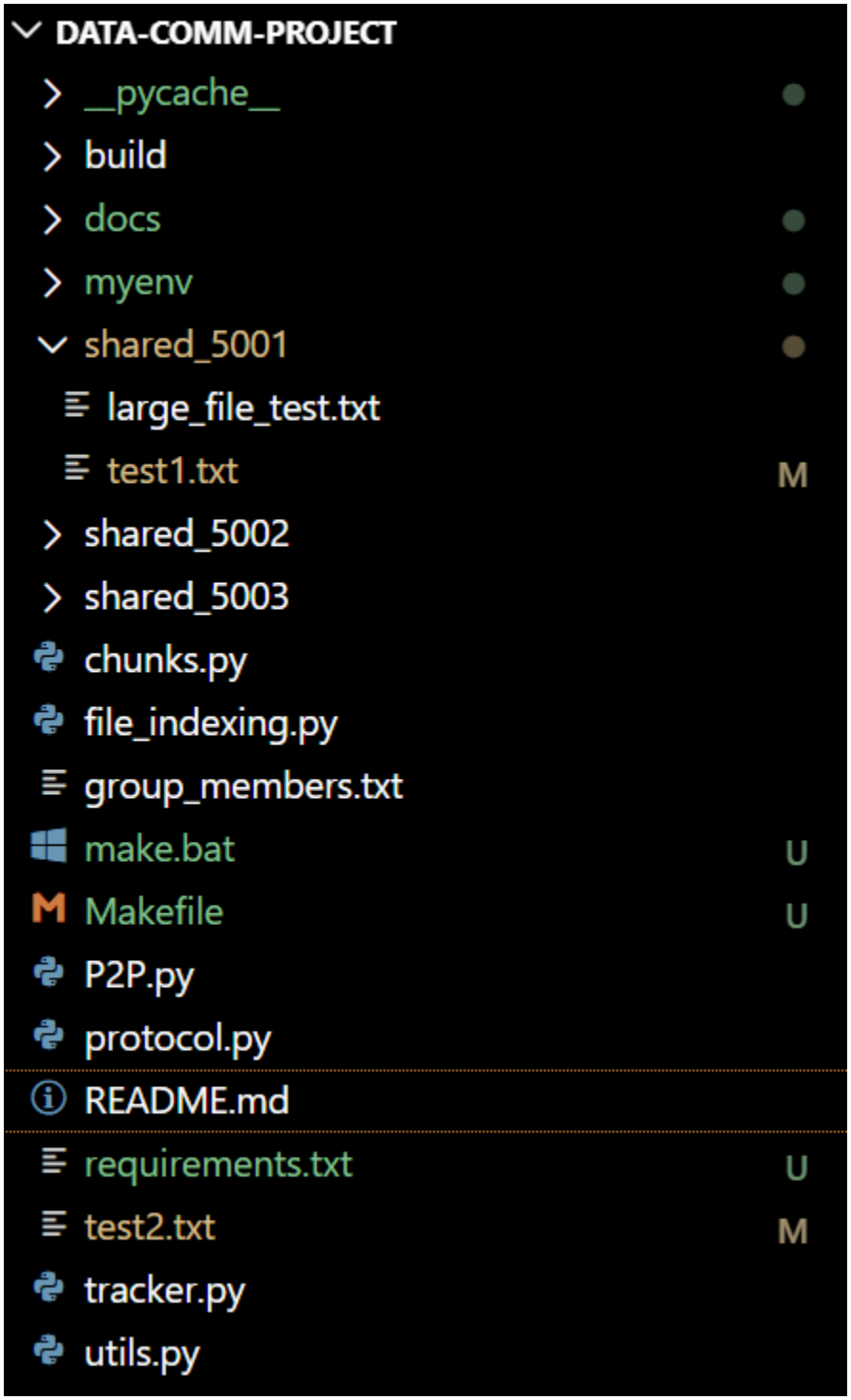
Project Structure:



Start Tracker:

```
(myenv) PS C:\Users\Elan\OneDrive\Documents\Data Comm\Project\Data-Comm-Project> python tracker
.py
[TRACKER RUNNING] 127.0.0.1:9000
```

Start multiple peers in separate terminals:

```
(myenv) PS C:\Users\Elan\OneDrive\Documents\Data Comm\Project\Data-Comm-Project> python P2P.py
5001
[LISTENING] Peer is listening on 127.0.0.1:5001
[TRACKER] REGISTERED 127.0.0.1:5001

>
```

```
(myenv) PS C:\Users\Elan\OneDrive\Documents\Data Comm\Project\Data-Comm-Project> python P2P.py
5002
[LISTENING] Peer is listening on 127.0.0.1:5002
[TRACKER] REGISTERED 127.0.0.1:5002

>
```

```
(myenv) PS C:\Users\Elan\OneDrive\Documents\Data Comm\Project\Data-Comm-Project> python P2P.py
5003
[LISTENING] Peer is listening on 127.0.0.1:5003
[TRACKER] REGISTERED 127.0.0.1:5003

>
```

Search for a file (.txt file located in shared_5002):

- 5001:

```
> search test1.txt
[INFO] Searching for 'test1.txt' across peers...
[FOUND] test1.txt at 127.0.0.1:5002

>
```

- 5002:

```
> [NEW CONNECTION] ('127.0.0.1', 55452) | Active: 1
[RECV ('127.0.0.1', 55452)] {'type': 'SEARCH', 'filename': 'test1.txt'}
[DISCONNECTED] ('127.0.0.1', 55452) | Active: 0
```

Search for a file that does not exist:

- 5001:

```
> search nonexistent_file.txt
[INFO] Searching for 'nonexistent_file.txt' across peers...
[NOT FOUND] nonexistent_file.txt not found in network

>
```

- 5002:

```
[NEW CONNECTION] ('127.0.0.1', 64033) | Active: 1
[RECV ('127.0.0.1', 64033)] {'type': 'SEARCH', 'filename': 'nonexistent_file.txt'}
[DISCONNECTED] ('127.0.0.1', 64033) | Active: 0
```

Send a message:

- 5001:

```
> echo 127.0.0.1 5003 hello?

>
```

- 5003:

```
[NEW CONNECTION] ('127.0.0.1', 63189) | Active: 1
[RECV ('127.0.0.1', 63189)] {'type': 'ECHO', 'message': 'hello?'}
[DISCONNECTED] ('127.0.0.1', 63189) | Active: 0
```

Verify connection:

- 5001:

```
> ping 127.0.0.1 5002
[PONG] from 5002

>
```

- 5002:

```
[NEW CONNECTION] ('127.0.0.1', 56240) | Active: 1
[RECV ('127.0.0.1', 56240)] {'type': 'PING'}
[DISCONNECTED] ('127.0.0.1', 56240) | Active: 0
```

Verify multiple connections:

- 5001:

```
> ping 127.0.0.1 5002
[PONG] from 5002

>
```

- 5003:

```
> ping 127.0.0.1 5002
[PONG] from 5002

>
```

- 5002:

```
[NEW CONNECTION] ('127.0.0.1', 62710) | Active: 1
[RECV ('127.0.0.1', 62710)] {'type': 'PING'}
[NEW CONNECTION] ('127.0.0.1', 62711) | Active: 2
[RECV ('127.0.0.1', 62711)] {'type': 'PING'}
[DISCONNECTED] ('127.0.0.1', 62710) | Active: 1
[DISCONNECTED] ('127.0.0.1', 62711) | Active: 0
```

Download file:

- 5001:

```
> download test1.txt 127.0.0.1 5002
[INFO] Downloading test1.txt (1 chunks)
[DOWNLOADED] chunk 0
[INFO] File saved to C:\Users\Elan\OneDrive\Documents\Data Comm\Project\Data-Comm-Project\shared_5001/test
1.txt
[SUCCESS] File integrity VERIFIED

>
```

- 5002:

```
[NEW CONNECTION] ('127.0.0.1', 52860) | Active: 1
[RECV ('127.0.0.1', 52860)] {'type': 'GET_CHUNK', 'filename': 'test1.txt', 'chunk': 0}
[DISCONNECTED] ('127.0.0.1', 52860) | Active: 0
```

Try to download a nonexistent file:

- 5001:

```
> download nonexistent_file.txt 127.0.0.1 5002
[ERROR] Could not get file info

>
```

- 5002:

```
[NEW CONNECTION] ('127.0.0.1', 56584) | Active: 1
[RECV ('127.0.0.1', 56584)] {'type': 'GET_FILE_INFO', 'filename': 'nonexistent_file.txt'}
[DISCONNECTED] ('127.0.0.1', 56584) | Active: 0
```

Download a large file:

- 5001:

```
> download large_file_test.txt 127.0.0.1 5002
[INFO] Downloading large_file_test.txt (489 chunks)
[DOWNLOADED] chunk 0
[DOWNLOADED] chunk 1
[DOWNLOADED] chunk 2
[DOWNLOADED] chunk 3
[DOWNLOADED] chunk 4
[DOWNLOADED] chunk 5
[DOWNLOADED] chunk 6
[DOWNLOADED] chunk 7
[DOWNLOADED] chunk 10
[DOWNLOADED] chunk 9
[DOWNLOADED] chunk 11
[DOWNLOADED] chunk 8
[DOWNLOADED] chunk 12
[DOWNLOADED] chunk 13
```

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```
[DOWNLOADED] chunk 460
[DOWNLOADED] chunk 462
[DOWNLOADED] chunk 463
[DOWNLOADED] chunk 465
[DOWNLOADED] chunk 464
[DOWNLOADED] chunk 466
[DOWNLOADED] chunk 467
[INFO] File saved to C:\Users\Elan\OneDrive\Documents\Data Comm\Project\Data-Comm-Project\shared_5001/large_file_test.txt
[SUCCESS] File integrity VERIFIED

>
```

Download a corrupted file:
NOTE: uncomment the following line in the function handle_client():

#If chunk_index == 2:

#Data = b”CORRUPTED_DATA”

- 5001:

```
> download large_file_test.txt 127.0.0.1 5002
[INFO] Downloading large_file_test.txt (489 chunks)
[DOWNLOADED] chunk 1
[DOWNLOADED] chunk 0
[DOWNLOADED] chunk 2
[DOWNLOADED] chunk 3
[DOWNLOADED] chunk 4
[DOWNLOADED] chunk 5
[DOWNLOADED] chunk 6
[DOWNLOADED] chunk 7
[DOWNLOADED] chunk 8
[DOWNLOADED] chunk 14
```

.
.
.

```
[DOWNLOADED] chunk 443
[DOWNLOADED] chunk 441
[DOWNLOADED] chunk 442
[DOWNLOADED] chunk 444
[DOWNLOADED] chunk 446
[DOWNLOADED] chunk 445
[DOWNLOADED] chunk 449
[DOWNLOADED] chunk 447
[DOWNLOADED] chunk 448
[INFO] File saved to C:\Users\Elan\OneDrive\Documents\Data Comm\Project\Data-Comm-Project\shared_5001/large_file_test.txt
[ERROR] File CORRUPTED

>
```

Maintain a connection:
NOTE: After establishing a connection, type ‘exit’ to lose the connection.

- 5001:

```
> connect 127.0.0.1 5002
[CONNECTED TO 127.0.0.1:5002]
> hello
{'type': 'ECHO_REPLY', 'message': 'hello'}
> how are you?
{'type': 'ECHO_REPLY', 'message': 'how are you?'}
```

- 5002:

```
[NEW CONNECTION] ('127.0.0.1', 49345) | Active: 1
[RECV ('127.0.0.1', 49345)] {'type': 'ECHO', 'message': 'hello'}
[RECV ('127.0.0.1', 49345)] {'type': 'ECHO', 'message': 'how are you?'}
```